

Motivation 4 — COMBSS in one page

Where it sits between lasso and MIO

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Draft outline — content to come.

The bridge

	Lasso	MIO	COMBSS
Variable type	$\beta \in \mathbb{R}^p$	$z \in \{0, 1\}^p$	$t \in [0, 1]^p$
Sparsity by	ℓ_1 penalty	hard $\ell_0 + \text{Big-}M$	continuous relaxation on the support indicator
Selection size	indexed by λ	k via constraint	k via polytope
Scales to $p \gg n$	yes	barely	yes
Biased coefficients	yes	no	no (envelope)

In one sentence

COMBSS relaxes the discrete support indicator $z \in \{0, 1\}^p$ to a continuous $t \in [0, 1]^p$ on the cardinality polytope, fits a ridge GLM in the relaxed space, and runs a Frank-Wolfe homotopy that drives t to a vertex of the polytope — recovering an integer subset of size exactly k .

Preview of the rest of the talk

- **Methodology** — how the relaxation works, the Frank-Wolfe step, the curvature homotopy.
- **Demos** — five runnable demos showing the method in action.

[← MIO](#) · [Methodology](#) → [Boolean relaxation](#)